Comparing long and short GRBs using Fermi catalog data

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ABSTRACT

Gamma-ray bursts are one of the most powerful events to happen in the universe. Knowing and understanding the entire population of GRBs can help humanity to understand the events to which GRBs are caused by. I present a study that seeks to compare the two variations of GRBs, long GRBs and short GRBs. I found the sky distribution of GRBs and the hardness verses log10[T90] plot. I received a $< sin^2(\delta) >$ value of 0.348 ± 0.007 for the long GRBs and a value of 0.348 ± 0.012 for the short GRBs. From the duration distribution plot (Figure 2), I found that the long GRB population have a likely duration of around 57.6 seconds whereas the short GRB population has a likely duration of around 1×10^{20} seconds.

1 INTRODUCTION

From the perspective of humans on Earth, space seems quiet and calm. However, it is very much not so. In the late 1960s, the Vela satellites caught a strange phenomenon that puzzled astronomers for decades (Kumar & Zhang 2015). This phenomenon was later called a gamma-ray burst. Gamma ray bursts (GRBs) are one of the most powerful events to occur in the universe. These events are comprised of powerful extragalactic gamma rays that are common with massive star death and binary compact object mergers (Luongo & Muccino 2021). They are also not widely considered standard candles, however, astronomers are working on the possibility of using correlations of GRB phenomenology to make GRBs into standard candles (Graziani 2011). To date, there are two variations of GRBs, long GRBs and short GRBs. Long GRBs have a timescale of more than 2 seconds and have been found to be one of the outcomes of massive star death (mass $\gtrsim 15M_{\odot}$) whereas short GRBs have a timescale of less than 2 seconds and are suspected as an outcome for mergers of compact objects in binary systems (neutron stars/black holes) (Kumar & Zhang 2015). In this paper, I present a study that seeks to compare these two variations of GRBs. The layout of the rest of this paper is as follows: in §2 we briefly describe the analysis method. The results of the analysis are given in §3. In §4, we discuss the results. Finally, in §5, we present our conclusions.

2 METHODS

First of all, I used the Fourth Fermi-GBM Gamma-Ray Burst Catalog (von Kienlin et al. 2020) for all the data in this analysis. Each analysis can be done using different programs and commands. In this paper, I used Google Colab (python). The first analysis is of the distribution of the long GRBs and the short GRBs on a mollweide plot. This analysis will show if there is any difference between the distributions of each variation along the sky. To plot the distributions, I take the right ascension and declination values from the catalog and use those to plot. The second analysis is the $< sin^2(\delta) >$ method where δ is the declination values. Higher values of $< sin^2(\delta) >$ mean a higher concentration of a population near the poles. I used the declination values from the catalog and the average function to get this value.

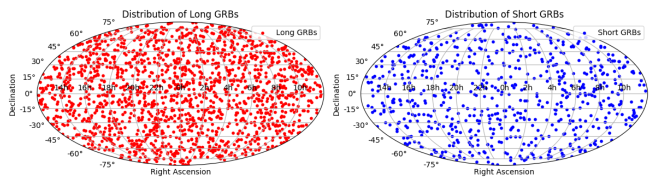

Figure 1. Figure 1: Mollweide plot of the distribution of long GRBs (red) and short GRBs (blue).

For the third analysis, I plot the durations distributions of the GRBs. I did this by using the timescale values (T90), taking the log10 of the values, and plotting them verses the frequency that they occur. I then used the unbinned maximum likelihood method for the line fit. For the forth analysis, I plotted the spectral hardness verses the log10[T90]. This is done by using

$$h = \frac{F_{10-1000keV}}{F_{50-300keV}} \tag{1}$$

where h is the hardness, $F_{10-1000keV}$ is the fluence in the 10-1000 keV range, and $F_{50-300keV}$ is the fluence in the 50-300 keV range. Take these values, take the log10 of those values, and plot them verses the log10[T90].

3 RESULTS

For the distribution of the GRBs , I received figure 1 . The long GRBs are represented by the red dots while the blue dots represent the short GRBs. For the $< sin^2(\delta) >$ analysis, I received a value of 0.348 ± 0.007 for the long GRBs and a value of 0.348 ± 0.012 for the short GRBs. For the plot of the duration distribution, I received figure 2. Using estimates on where the line fit rate of change is 0, I found that the long GRB population has a likely duration of around 57.6 seconds whereas the short GRB population has a likely duration of around 1×10^{20} seconds. For the plot of the spectral hardness verses the log10[T90], I received figure 3.

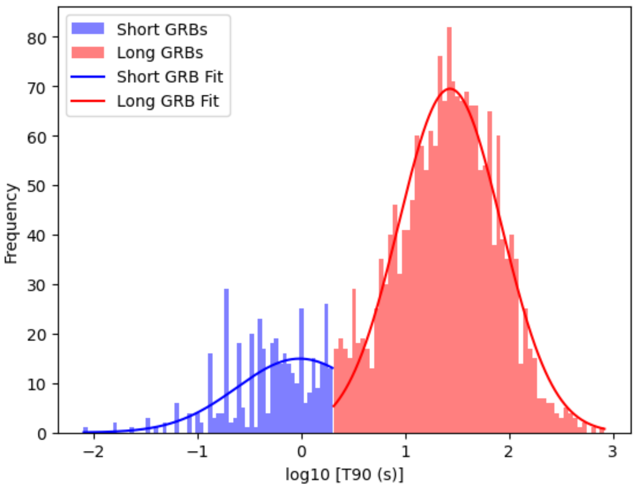

Figure 2. Figure 2: plot of the frequency vs. the log10[T90].

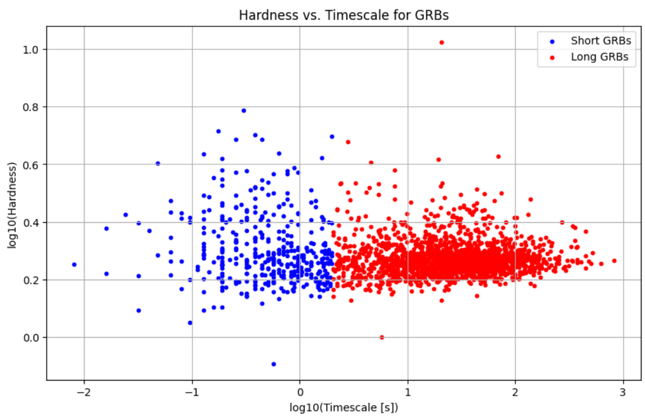

Figure 3. Figure 3: plot of the spectral hardness verses the timescale on a log10 fit.

4 DISCUSSION

4.1 Figure 1 analysis

The mollweide projection of the GRB distribution reveal the properties of the GRB location on the sky. In comparison between the short and long GRBs, there are no outstanding differences between the distribution. However, there is an outstanding difference between the number of long and short GRBs shown. In Figure 1, there are many more long GRBs than there are short GRBs. This can mean that in the time-frame of this study, there were more events captured that were associated with long GRBs rather with short GRBs. This leads to the conclusion that the events associated with short GRBs are more rare in context.

4.2 $< sin^2(\delta) >$ analysis

The $< sin^2(\delta) >$ analysis resulted in values that revealed that the long and short GRBs do not have outstanding differences in concentration at the poles of Figure 1. This makes sense for the fact that the long and short GRBs have about the same distribution. In a broader sense of analysis, this means that no part in the sky distribution has an outstanding significance compared to the rest of the distribution.

4.3 Figure 2 analysis

Figure 2 represents the duration distribution of the GRB population. This distribution is bimodal with two distinct peaks. The unbinned maximum likelihood method fit reveals each GRB variation most likely duration wherever the fit peaks. These values are the most likely, however, there are other values other than these in the population. The results from figure 2 reveals that even through the two GRB classifications are separated by the 2 second threshold, the most likely values of duration in these populations are vastly different from each other. These values can further astronomy research in GRBs by studying the type of event that these likely durations originate from.

4.4 Figure 3 analysis

Figure 3 represents the plot spectral hardness verses log10[T90]. The long GRB population (represented by the red dots) seem to spread out and soften as timescale increases. The short GRB population (represented by the blue dots) exhibit a more harder spectrum. In terms, the short GRB population seems to have more intense bursts compared to their long counterparts where the long GRB population as a softer hardness. This makes sense, due to short GRBs known to be released from more intense events like object mergers whereas long GRBs are known to come from massive star collapse (still very intense but not when compared to short GRBs).

5 CONCLUSIONS

I have found that the short GRB population and the long GRB population have similarities but also obvious differences. Both of the long and short GRB populations have similar sky distributions and $< sin^2(\delta) >$ values, thus no outstanding evidence to suggest that the long or short GRB population prefer a section of the sky. On the contrary, the short GRB population has a very small likely duration with a more intense hardness where the long GRB population has a large likely duration and exhibits a softer hardness. Future work into the extremes of both these GRB populations needs to be done in order to know the full spectrum of how these bursts form and by which way they form.

REFERENCES

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